

06 The reaction quotient, the equilibrium constant, and rate laws for reversible reactions

These notes are based off [this chapter](#) and [this chapter](#) from ChemLibre Texts.

Let's start with the equilibrium constant. In this class we denote K_c is the

equilibrium constant in terms of concentration and K_p is the equilibrium constant

in terms of pressure.

Note that K_p is only valid for gases, since pressure in a liquid is not well-defined

For a chemical reaction, $aA + bB \rightarrow cC + dD$, the condition for equilibrium is

chemical potential of reactants = chemical potential of products

Therefore $a\mu_A + b\mu_B = c\mu_C + d\mu_D$

Where μ are chemical potentials

$$\mu = \mu^0 + RT \ln a$$

Where μ^0 is the chemical potential at standard state

And a is the activity coefficient

For an ideal substance, $\mu^0 = \Delta G_f^0$ (Gibbs Free energy of formation)

And $a = \text{Concentration (C)}$

Hence, $a\mu_A + b\mu_B = c\mu_C + d\mu_D$ can be re-written

$$a\Delta G_{f,A}^0 + aRT \ln C_{A,\text{eq}} + b\Delta G_{f,B}^0 + bRT \ln C_{B,\text{eq}} = c\Delta G_{f,C}^0 + cRT \ln C_{C,\text{eq}} + d\Delta G_{f,D}^0 + dRT \ln C_{D,\text{eq}}$$

Where the subscript 'eq' indicates these are the equilibrium concentrations

This equation can be re-arranged:

$$a\Delta G_{f,A}^0 + b\Delta G_{f,B}^0 - c\Delta G_{f,C}^0 - d\Delta G_{f,D}^0 = RT \ln \frac{C_{C,\text{eq}}^c C_{D,\text{eq}}^d}{C_{A,\text{eq}}^a C_{B,\text{eq}}^b}$$

Where $a\Delta G_{f,A}^0 + b\Delta G_{f,B}^0 - c\Delta G_{f,C}^0 - d\Delta G_{f,D}^0 = -\Delta G_{\text{rxn}}$ (Gibbs free energy of reaction)

And $\frac{C_{C,\text{eq}}^c C_{D,\text{eq}}^d}{C_{A,\text{eq}}^a C_{B,\text{eq}}^b} = K_c$ (equilibrium constant)

$$\text{Hence, } K_c(T) = \frac{C_{C,\text{eq}}^c C_{D,\text{eq}}^d}{C_{A,\text{eq}}^a C_{B,\text{eq}}^b} = \exp\left(-\frac{\Delta G_{\text{rxn}}}{RT}\right)$$

$$\Delta G_{rxn}(T) = \Delta H_{rxn} - T \Delta S_{rxn}$$

$$\text{Where } \Delta H_{rxn} = d\Delta H_{f,D}^0 + c\Delta H_{f,C}^0 - a\Delta H_{f,A}^0 - b\Delta H_{f,B}^0$$

$$\text{And } \Delta S_{rxn} = dS_D + cS_C - aS_A - bS_B$$

Values of ΔH_f^0 and S can be looked up, e.g., using [the NIST Chemistry Webbook](#).

The reaction quotient, $Q = \frac{c_C^c c_D^d}{c_A^a c_B^b}$ which is related to K_c EXCEPT THAT Concentrations are
not their equilibrium values

The quantity $\frac{Q}{K_c}$ is known as the "reversibility" and tells us something about the approach
to equilibrium

Specifically, if $\frac{Q}{K_c} > 1$, reactants are favored, rxn runs in reverse

And if $\frac{Q}{K_c} < 1$, products favored, rxn runs in forward

We can use these quantities to write rate laws for reversible reactions. Consider the reaction from above: $aA + bB \rightleftharpoons cC + dD$.

The rate law is $-r_A = k_{f,A} C_A^a C_B^b - k_{r,A} C_C^c C_D^d$

Where k_f = forward rate constant $= A_f \exp\left(-\frac{\Delta G_f^{\text{act}}}{RT}\right)$

we will show later that $k_f = \frac{k_B T}{h} \exp\left(-\frac{\Delta G_f^{\text{act}}}{RT}\right)$

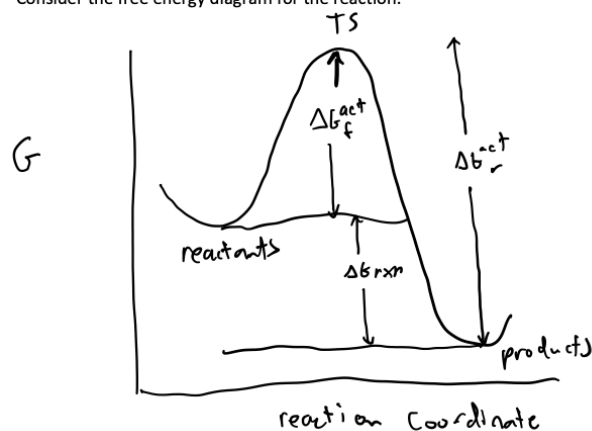
k_B = Boltzmann constant $= \frac{R}{N_A}$
 h = Planck constant

And k_r =

$$\frac{k_B T}{h} \exp\left(-\frac{\Delta G_r^{\text{act}}}{RT}\right)$$

How are k_f and k_r related?

Consider the free energy diagram for the reaction:



note how

$$\Delta G_r^{\text{act}} = \Delta G_f^{\text{act}} - \Delta G_{\text{rxn}}$$

Hence, $\frac{k_f}{k_r} = \frac{\exp\left(-\frac{\Delta G_f^{\text{act}}}{RT}\right)}{\exp\left(-\frac{\Delta G_r^{\text{act}}}{RT}\right)} = \exp\left(\frac{-\Delta G_f^{\text{act}} + \Delta G_r^{\text{act}}}{RT}\right) = \exp\left(-\frac{\Delta G_{\text{rxn}}}{RT}\right) = K_c$

~~Assuming $\Delta G_{\text{rxn}} = \Delta G_f^{\text{act}} - \Delta G_r^{\text{act}}$~~

$$-r_A = k_f C_A^a C_B^b - k_r C_C^c C_D^d$$

↗ factor out k_f get this equation

Hence, we can re-write the rate law as $-r_A = k_f \left(C_A^a C_B^b - \frac{C_C^c C_D^d}{K_c} \right)$, which is the form that we normally use.

In terms of the reaction quotient: $-r_A = k_f C_A^a C_B^b \left(1 - \frac{Q}{K_c} \right)$. At equilibrium $Q = K_c$ hence

$$\boxed{-r_A = 0 \text{ @ equilibrium}}$$

Temperature dependence of the equilibrium constant:

$$K_c(T) = \exp\left(-\frac{\Delta G_{\text{rxn}}}{RT}\right) = \exp\left(-\frac{\Delta H_{\text{rxn}} - T\Delta S_{\text{rxn}}}{RT}\right) = \exp\left(-\frac{\Delta H_{\text{rxn}}}{RT}\right) \exp\left(\frac{\Delta S_{\text{rxn}}}{R}\right)$$

Assuming ΔH_{rxn} and ΔS_{rxn} are constant with temperature,

$$\frac{K_c(T_2)}{K_c(T_1)} = \frac{\exp\left(-\frac{\Delta H_{\text{rxn}}}{RT_2}\right)}{\exp\left(-\frac{\Delta H_{\text{rxn}}}{RT_1}\right)} = \exp\left[\frac{-\Delta H_{\text{rxn}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

$$\text{So } K_c(T_2) = K_c(T_1) \exp\left[\frac{-\Delta H_{\text{rxn}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

this is the van t Hoff equation